

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	ABCT3114
Subject Title	Biochemical Techniques
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Biochemistry
Objectives	Through lectures and tutorials, this subject is intended to provide students with an understanding of the fundamental principles of instruments and techniques commonly used in biochemistry and molecular biology.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Gain a comprehensive understanding of the principles and theories of various techniques on purifying and analyzing biomolecules and be able to design/criticize protocol. - Develop the ability to critically analyze and interpret biochemical data presented in scientific literature, understanding the strengths and limitations of various techniques and how they contribute to advancements in biochemical research. - Apply knowledge of various biological techniques including fluorescence, luminescence, real-time PCR, gel electrophoresis to solve theoretical problems in modern day biological research projects.
Subject Synopsis/ Indicative Syllabus	- Principles & practices of chromatography – permeation, ionexchange, affinity chromatography; gas chromatography; HPLC; separation of biomolecules - Electrophoresis: principles and general techniques; special techniques-high voltage discontinuous, SDS-polyacrylamide, gradient, isoelectric focusing, two-dimensional electrophoresis and western blotting - Fluorescence principle and its applications; flow cytometry - DNA microarray - DNA sequencing and fingerprinting - Luminescence and chemiluminescence; luciferase, green fluorescent protein

	• Real-time PCR • Sample preparation: cell disruption, buffer, dialysis, concentration methods, measurement of proteins and nucleic acids • Centrifugation and subcellular fractionation																																															
Teaching/Learning Methodology	Lectures will be used to introduce the basic principles. Emphasis is put on the application of these techniques, rather than memorizing the detailed principles. Exercises and case studies in tutorials are used to teach students how to analyze experimental data and design experiments to solve some basic biochemical problem.																																															
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th></th><th></th></tr><tr><td>1. Assignments</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>2. Mid-term Quiz</td><td>30%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td colspan="5"></td></tr></table> Assignments will include exercises and case studies in tutorials. We will test the ability of students to understand the principles behind different biochemical techniques and assess their technical skills. The exercises and case studies in tutorials and preparatory reports will require students to understand the principles and applying these principles to analyse experimental results. There will be one mid-term quiz to test the students’ understanding of basic principles. Experimental results will be presented in the quiz and students are expected to interpret the experimental data and be able to propose experiments to solve some basic biochemical problems. Examinations will be of the style of analysing experimental results and proposing new experiments based on the preliminary results provided in the questions. There is no need for memorizing difficult principles as such part will be a relatively small part of the examination. Students are required to attend at least 75% of scheduled sessions for the subject. Students fail to fulfill the attendance requirement will not be eligible to register Molecular Diagnostics in Pathology Specialties (ABCT4108).	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c			1. Assignments	20%	✓	✓	✓			2. Mid-term Quiz	30%	✓	✓	✓			3. Examination	50%	✓	✓	✓			Total	100%												
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Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial	13 Hrs.
	Other student study effort:	
	▪ Self-study	60 Hrs.
	▪ Assignment	10 Hrs.
	Total student study effort	109 Hrs.
Reading List and References	Text book: Statistics. Donnelly RA Jr and Abdel-Raouf F. 2016. Alpha, Indianapolis, Indiana. ISBN : 9781465454096 Recommended reading: The lady tasting tea : how statistics revolutionized science in the twentieth century. Salsburg D. 2001. Freeman, New York, New York. ISBN : 0716741067 Library location: Q175.S2345 2001 Hands-on exploratory data analysis with R: Become an expert in exploratory data analysis using R packages. Datar R, Garg H. 2019. Packt Publishing, Birmingham, UK. ISBN: 178980437X Library online access	